THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF DEFENCE AND NATIONAL SERVICE
MILITARY SCHOOLS ASSOCIATION
FORM TWO PRE-MOCK JUNE 2026

MARKING SCHEME

COMPUTER SCIENCE 1

SECTION A (15 Marks)

Answer all questions in this section.

QUESTION 1 (10 Marks @ 1 mark each)

Each multiple-choice question carries 1 mark.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
D	B	B	B	A	C	B	D	A	D

QUESTION 2 (5 Marks @ 1 mark each)

Each correct match carries 1 mark. (Options A and F are distractors and are not used.)

(i)	(ii)	(iii)	(iv)	(v)
C	G	D	B	E

SECTION B (60 Marks)

Note: The original question paper's numbering jumps directly from Question 5 to Question 7 — Question 6 does not appear in the source paper. This marking scheme follows the question numbers exactly as given.

Answer all questions in this section.

QUESTION 3 (10 Marks)

(a) Explain the role of computer science in developing practical solutions to enhance transportation systems. (5 marks)

- GPS and navigation systems help plan routes, track locations, and provide real-time traffic updates.
- Ride-hailing/ride-sharing mobile apps (e.g. Uber, Bolt) connect drivers and passengers efficiently.
- Smart traffic management systems use sensors and software to control traffic lights and ease congestion.
- Vehicle tracking and fleet management systems allow monitoring of vehicles for safety and efficiency.
- Development of autonomous/self-driving vehicle technology using artificial intelligence and sensors.
- (Accept: electronic/contactless ticketing systems for public transport.)

o 1 mark for each valid, well-stated point, up to a maximum of 5 marks.

(b) Discuss how computer science improves teaching and learning, and give examples of software/platforms that support education delivery. (5 marks)

- E-learning platforms (e.g. Google Classroom, Moodle) allow students to access lessons and materials remotely.
- Multimedia software (videos, simulations, animations) makes lessons more interactive and easier to understand.
- Online assessment tools (e.g. Kahoot, Google Forms) allow quick, automated testing and feedback.
- Communication platforms (e.g. Zoom, WhatsApp, e-mail) enable interaction between teachers and students beyond the classroom.

• Digital libraries and online research databases give access to a wide range of learning resources.

o 1 mark for each valid point that names a benefit and/or a supporting software/platform, up to a maximum of 5 marks.

QUESTION 4 (10 Marks)

(a) Why the first generation of computers were large, overheating, and power inefficient. (3 marks)

Answer: First generation computers used vacuum tubes as their main electronic component. Vacuum tubes are physically bulky, generate a great deal of heat during operation, and consume large amounts of electrical power — which together made these computers large in size, prone to overheating, and inefficient in their use of power.

o 3 marks: Correctly identifies vacuum tube technology and links it to all three features (size, heat, power).

o 2 marks: Identifies vacuum tubes and explains one or two of the features.

o 1 mark: Vague answer with little linkage to the cause.

(b) Why the development of microprocessors was a major development in computer technology. (3 marks)

Answer: The microprocessor integrated the entire processing unit (control unit and ALU) onto a single small chip. This made computers much smaller, cheaper, faster, and more power-efficient, and made possible the mass production of personal computers and other digital devices.

o 3 marks: Explains integration onto a single chip and the resulting benefits.

o 2 marks: States some benefits without explaining the integration.

o 1 mark: A vague or incomplete answer.

(c) The role of the motherboard in a computer system and how it connects the computer's components. (4 marks)

Answer: The motherboard is the main circuit board of the computer. It houses and interconnects the major components — the CPU, RAM, storage devices, expansion cards and input/output ports — through built-in buses, sockets, and slots, allowing all these components to communicate with one another and work together as a single system.

o 4 marks: A complete explanation covering the motherboard's role as the main circuit board and how it interconnects components via buses/sockets/slots.

o 2-3 marks: A partial explanation covering only some of these points.

o 1 mark: A vague definition only.

QUESTION 5 (8 Marks)

(a) Two differences between primary and secondary memory. (2 marks)

• Primary memory (e.g. RAM) is volatile — it loses its data when the power is switched off — while secondary memory (e.g. HDD/SSD) is non-volatile and retains data permanently.

• Primary memory is faster and is directly accessed by the CPU during processing, while secondary memory is slower and is mainly used for long-term/bulk storage of data and programs.

o 1 mark for each correctly stated difference, maximum 2 marks.

(b) Steps to create a folder on the desktop. (6 marks)

• Right-click on an empty area of the desktop.

• From the shortcut menu that appears, point to/click on "New".

• Click on "Folder" from the submenu that appears.

• A new folder icon appears on the desktop with a default name highlighted/selected.

• Type the desired name for the folder.

• Press "Enter" (or click elsewhere on the desktop) to save and confirm the folder name.

o 1 mark for each correct step given in a logical order, maximum 6 marks.

QUESTION 7 (12 Marks)

(a) Flowchart and pseudocode for finding the area of a square foundation. (10 marks — 5 for the

flowchart, 5 for the pseudocode)

Pseudocode:

START
INPUT side
Area = side × side
DISPLAY Area
STOP

Flowchart (symbols in sequence, connected by arrows):

• Start/Terminal symbol (oval) — "Start"
• Input/Output symbol (parallelogram) — "Input side"
• Process symbol (rectangle) — "Area = side × side"
• Input/Output symbol (parallelogram) — "Display Area"
• Start/Terminal symbol (oval) — "Stop"

o 5 marks (pseudocode): All 5 logical steps present, in the correct order, using clear instructions.
o 5 marks (flowchart): All steps represented with the correct standard symbols, in the correct order, with arrows correctly connecting them.
o Award partial marks (3-4) for a mostly correct answer with one minor omission, and 1-2 marks for a weak attempt.

(b) Explain the terms 'flowchart' and 'pseudocode'. (2 marks)

• Flowchart: A diagrammatic/graphical representation of the sequence of steps in an algorithm, using standard symbols (e.g. oval, rectangle, parallelogram, diamond) connected by arrows to show the flow of execution.
• Pseudocode: A plain-language, structured description of an algorithm's logic, written in a way that resembles programming code but is not tied to the syntax of any specific programming language.
o 1 mark for each correctly explained term.

QUESTION 8 (10 Marks)

(a) The software that should be installed on a computer system before any other program. (2 marks)

Answer: The Operating System (e.g. Windows, Linux, or macOS). It must be installed first because it manages the computer's hardware resources and provides the platform on which all other application software runs.

o 2 marks: Correctly names the Operating System and briefly explains why it comes first.
o 1 mark: Names the Operating System only, without explanation.

(b) Four functions of the software named in 8(a). (8 marks)

• Manages the computer's hardware resources (CPU, memory, storage devices, and peripherals).
• Provides a user interface (graphical or command-line) through which the user interacts with the computer.
• Manages files and folders (creating, storing, retrieving, and organising data).
• Runs and manages other application programs, allocating memory and processor time to them.
o 2 marks for each correctly stated function, maximum 8 marks. (Accept other valid functions such as managing user accounts/security or handling multitasking.)

QUESTION 9 (10 Marks)

(a) Difference between the Internet and a website. (2 marks)

Answer: The Internet is a vast global network that connects millions of computers and devices worldwide, allowing them to communicate and exchange data. A website, on the other hand, is a collection of related web pages containing text, images and other content, which is hosted on a server and accessed through the Internet using a web browser. In short, the Internet is the underlying network/infrastructure, while a website is a service/content that runs on that infrastructure.

o 2 marks: Both terms clearly defined and correctly distinguished.

o 1 mark: Only one term correctly defined, or a vague distinction.

(b) Four reasons for computer programming. (8 marks)

• To create software applications that solve specific real-world problems.
• To automate repetitive tasks, saving time and reducing human error.
• To develop systems that improve efficiency in different sectors (e.g. education, health, business).
• To enable computers to perform calculations and process data according to given instructions.
o 2 marks for each valid reason, maximum 8 marks.

SECTION C (15 Marks)

QUESTION 10 (15 Marks)

Explain the advantages and disadvantages of using electronic mail. (3 points)

Advantages:

• Fast and instant delivery of messages regardless of the distance between sender and receiver.
• Cost-effective compared to postal mail — no need for paper, stamps, or physical delivery.
• Allows attachments (documents, images, and other files) to be sent along with the message.

Disadvantages:

• Vulnerable to hacking, phishing, spam, and malware.
• Requires internet access, electricity, and a working device to send or receive messages.
• Messages can be misdirected, ignored, or lost in a crowded inbox, and lack the personal touch of face-to-face communication.
o 5 marks for each well-explained point, up to a maximum of 3 points (15 marks total).
o Points may be any combination of advantages and disadvantages.
o Award partial marks for a point that is identified but not well explained.
o Accept any other plausible, well-argued advantage or disadvantage.